

Manufacturing Processes

Casting • Machining • Welding • Forming • Joining • Finishing

What You Will Learn

- Four categories of manufacturing: shaping, machining, joining, and finishing
- Primary shaping: casting, forging, rolling, extrusion — how each works
- Machining operations: turning, milling, drilling, grinding — tools and uses
- Welding processes (MIG, TIG, SMAW), heat treatment, and process selection

Process Selection Decision Criteria

Quantity?	Low = machining/casting High = stamping/injection moulding
Tolerance?	Tight = machining/grinding Loose = casting/forging
Material?	Steel = machining/welding Polymer = injection moulding/extrusion

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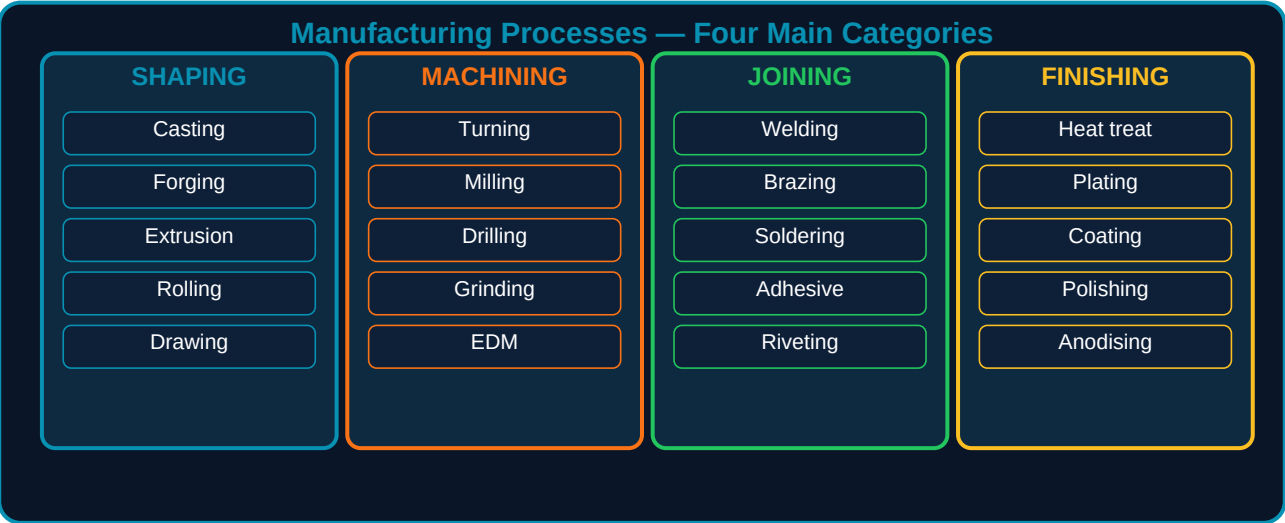
MECH-008: Manufacturing Processes

Every physical engineering product — a shaft, a pump housing, a bracket, a pipe fitting — begins as raw material and is transformed into its final shape through one or more manufacturing processes. The selection of the right manufacturing process is a fundamental engineering decision that affects product cost, quality, dimensional accuracy, surface finish, material properties, and production rate. There is no universally "best" process — only the best process for a given combination of material, geometry, quantity, and quality requirements.

Learning Objectives

- Categorise manufacturing processes into the four main groups: shaping, machining, joining, and finishing
- Describe the key primary shaping processes: sand casting, die casting, hot forging, rolling, and extrusion
- Explain turning, milling, drilling, and grinding operations with their tools and applications
- Compare MIG, TIG, and SMAW welding processes and select the appropriate process for given conditions
- Apply process selection criteria (material, quantity, tolerance, geometry) to choose the right manufacturing method

1. The Four Categories of Manufacturing



Category	What It Does	Key Characteristic	Examples
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Shaping (primary)	Transforms raw material into a near-final shape. Material is not removed — it is redistributed or solidified.	No material removal. Often produces near-net shapes with minimal waste. Lower tolerance than machining.	Sand casting, die casting, hot forging, rolling, extrusion, injection moulding
Machining (subtractive)	Removes material from a workpiece using cutting tools to achieve precise dimensions and surface finish.	Highest dimensional accuracy and surface quality. Material wasted as chips. Most flexible for complex shapes.	Turning, milling, drilling, grinding, boring, EDM, laser cutting
Joining	Connects two or more components permanently or semi-permanently.	Enables manufacture of complex assemblies from simpler components. Permanent or releasable.	Welding (MIG/TIG/SMAW), brazing, soldering, riveting, adhesive bonding, bolting
Finishing (surface)	Improves surface properties: hardness, wear resistance, corrosion protection, aesthetics.	Applied after shaping and machining. Changes surface without significant dimensional change.	Heat treatment, electroplating, anodising, painting, polishing, coating, nitriding

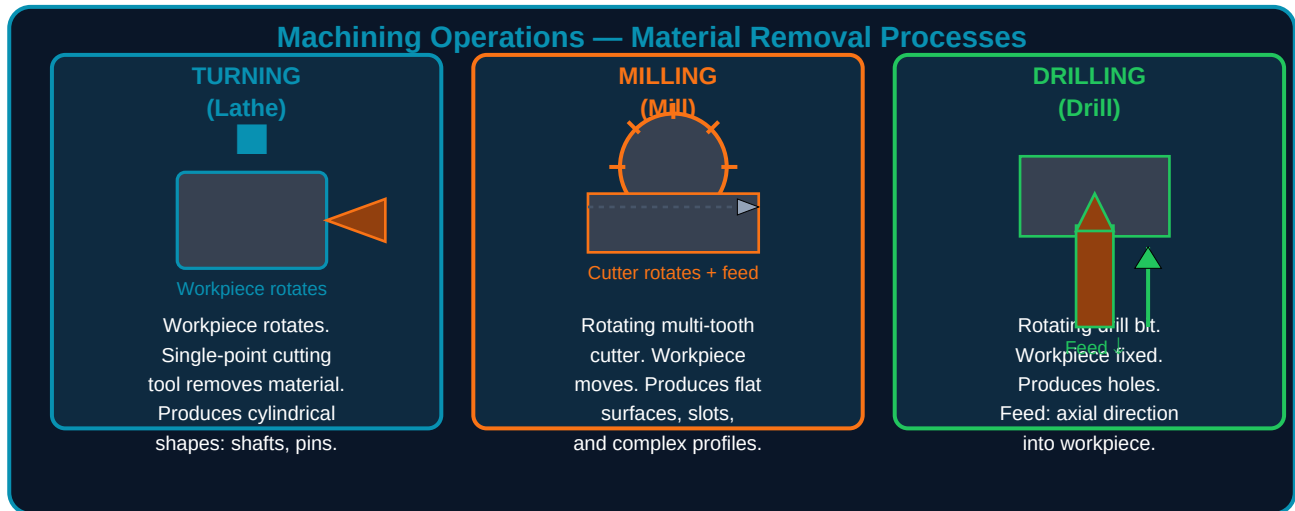
2. Primary Shaping Processes

Primary shaping processes produce the initial form of a component from raw material. The choice of shaping process determines the basic economics of manufacturing — tooling cost, production rate, material utilisation, and the dimensional accuracy achievable before secondary machining.

Process	How It Works	Best For	Limitations
Sand Casting	Molten metal poured into a sand mould. Mould destroyed to extract part. New mould for each cast.	Large, complex shapes. Low-to-medium volume. Wide range of metals including iron, aluminium, steel.	Poor surface finish (Ra 12–25 μm). Loose tolerances ($\pm 0.5\text{--}3\text{mm}$). Porosity risk in cast structure.
Die Casting	Molten metal injected under high pressure into a permanent steel die. Very fast cycle.	High volume production of non-ferrous parts (aluminium, zinc, magnesium). Automotive housings, consumer electronics.	High tooling cost (\$10,000–\$100,000+). Only for non-ferrous metals. Limited wall thickness variation.
Hot Forging	Metal heated above recrystallisation temperature and hammered/pressed into shape using dies.	High-strength structural parts: crankshafts, gears, connecting rods, flanges. Better mechanical properties than casting.	High tooling cost. Limited to simpler shapes than casting. Requires secondary machining for tight tolerances.
Rolling	Metal passed between rotating rollers to reduce thickness and produce flat or shaped sections.	Structural steel sections (I-beams, channels, angles, plate). Wire, rod, and sheet production. Very high volume.	Limited to constant cross-section shapes. Cannot produce complex 3D geometries in a single pass.

Extrusion	Material (hot metal or plastic) forced through a shaped die opening to produce constant cross-section profiles.	Aluminium window/door sections, plastic pipes, complex hollow profiles. Long lengths of constant section.	Limited to constant cross-section. Die cost significant for complex profiles. Metals: mainly Al, Cu, Pb, Mg.
Injection Moulding	Molten polymer injected into a closed mould under high pressure. Mould opens to eject solid part.	Very high volume plastic parts. Automotive trim, housings, containers. Excellent repeatability.	Very high tooling cost (\$20,000–\$500,000). Only for polymers and composites. Long lead time for tooling.

3. Machining — Precision Material Removal



Key Machining Operations

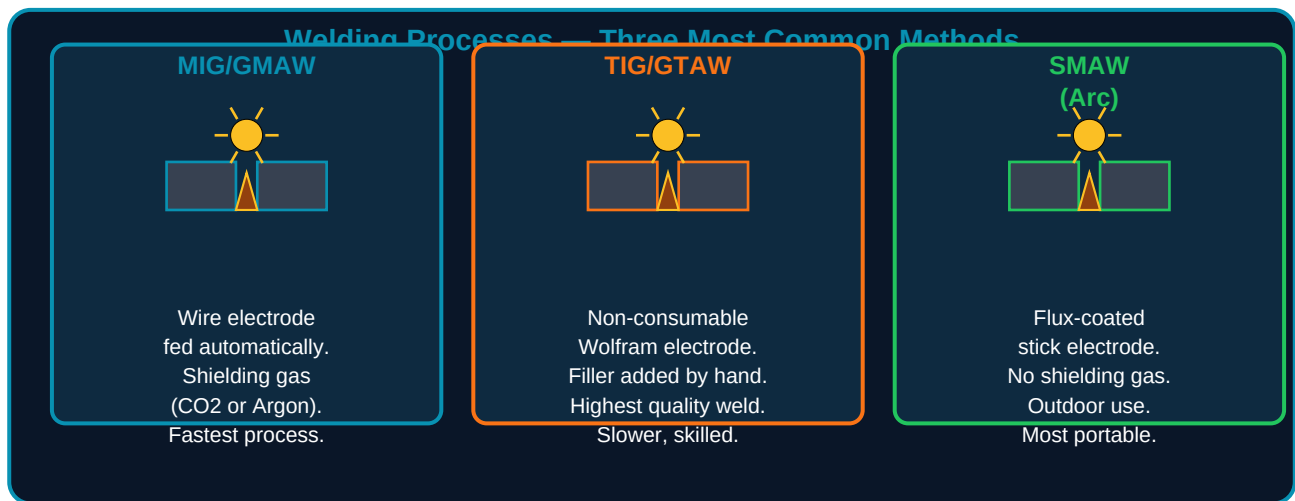
Operation	Machine	Cutting Motion	What It Produces	Typical Tolerance
Turning	Lathe (CNC or manual)	Workpiece rotates; single-point tool feeds	Cylinders, cones, threads, grooves, tapers, bores	$\pm 0.01\text{--}0.05\text{ mm}$
Milling	Milling machine (CNC)	Multi-tooth rotating cutter; workpiece feeds in X/Y/Z	Flat surfaces, slots, pockets, profiles, contours, gear teeth	$\pm 0.01\text{--}0.1\text{ mm}$
Drilling	Drill press, CNC machining centre	Drill rotates and feeds axially into workpiece	Holes (through or blind). Reaming follows for precision holes.	$\pm 0.05\text{--}0.2\text{ mm}$ (drill); $\pm 0.005\text{ mm}$ (reamed)
Grinding	Surface/cylindrical grinder	Abrasive wheel rotates at high speed; fine feed	Very flat surfaces, precise cylindrical dimensions, fine surface finish	$\pm 0.001\text{--}0.01\text{ mm}$

Boring	Boring machine, lathe, machining centre	Single-point tool rotates inside existing hole	Enlarges and corrects existing holes to precise diameter and roundness	$\pm 0.005\text{--}0.02\text{ mm}$
EDM (Electrical Discharge)	EDM machine	Electrical sparks erode material — no mechanical contact	Complex 3D cavities, hardened steels, die and mould making	$\pm 0.005\text{--}0.05\text{ mm}$

Cutting Tool Materials — Selection Guide

Tool Material	Hardness	Max Cutting Speed	Best For
High Speed Steel (HSS)	HRC 62–65	Low-medium (20–50 m/min)	Soft materials, complex tool shapes, low production volume. Still common for drills and taps.
Cemented Carbide (WC-Co)	HRA 88–93	High (100–400 m/min)	Steel, cast iron, stainless. Most common insert material in CNC machining. Indexable inserts.
Coated Carbide (TiN, TiAlN)	Surface HV 2000+	Very high (200–600 m/min)	Same as carbide but higher speed and tool life. Standard for production CNC machining.
Ceramic (Al ₂ O ₃ , Si ₃ N ₄)	HV 1800–3000	Very high (300–800 m/min)	Cast iron, hardened steel at high speed. No coolant. Brittle — not for interrupted cuts.
CBN (Cubic Boron Nitride)	HV 4000–4500	Very high (100–600 m/min)	Hardened steel (HRC 45+), turning hard materials. Expensive — used where nothing else works.
PCD (Polycrystalline Diamond)	HV 6000–8000	Highest (200–2000 m/min)	Non-ferrous (aluminium, copper, composites). Cannot machine steel — diamond reacts with iron.

4. Welding and Joining Processes



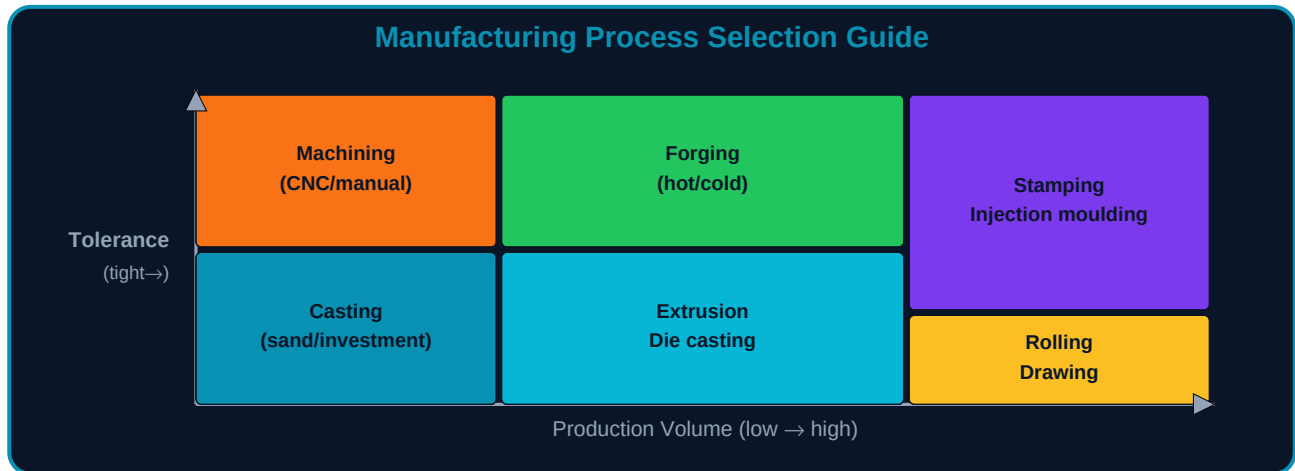
Process	Heat Source	Electrode / Filler	Shielding	Best Applications
MIG/GMAW	Electric arc	Continuous wire electrode (consumable)	Inert/active gas (CO ₂ , Ar, mix)	General fabrication, structural steel, automotive. Fast, easy to learn, semi-automatic.
TIG/GTAW	Electric arc	Tungsten electrode (non-consumable). Filler rod added separately by hand.	Argon or helium	Stainless steel, aluminium, thin materials, pressure vessels. Highest quality, slowest, most skilled.
SMAW (Stick)	Electric arc	Flux-coated stick electrode (consumable)	Flux coating generates shielding gas	Site work, field repairs, structural steel. Most portable. Works outdoors and in contaminated conditions.
Oxy-Acetylene	Combustion flame	Filler rod added by hand	None (or flux)	Thin sheet, pipe work, brazing, cutting. Low cost equipment. Not for structural steelwork.
Submerged Arc (SAW)	Electric arc under flux blanket	Continuous wire electrode	Granular flux blanket	Heavy plate, pressure vessels, pipe manufacture. Very high deposition rate. Automatic process only.

5. Surface Finishing and Heat Treatment

Process	What It Does	Improves	Common Application
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Quench and Temper	Heat to austenite range, quench rapidly in water/oil, then temper to reduce brittleness.	Hardness and strength. Temper controls toughness vs hardness trade-off.	Structural steel components, tools, shafts requiring high strength
Case Hardening (Carburising)	Diffuse carbon into surface layer of low-carbon steel, then harden surface.	Surface hardness and wear resistance while maintaining tough core.	Gear teeth, camshafts, bearing races — hard surface, tough core
Nitriding	Diffuse nitrogen into steel surface at low temperature (500–600°C). No quench needed.	Surface hardness without distortion. Excellent fatigue resistance.	Precision components (thread gauges, dies) that cannot be reground after heat treatment
Electroplating	Deposit thin metal layer (chrome, nickel, zinc) on substrate using electrolytic process.	Corrosion resistance, appearance, hardness, wear resistance.	Chrome plating on hydraulic cylinders, zinc plating on fasteners, nickel for electronics
Anodising	Electrolytic oxidation of aluminium surface to form thick, hard Al ₂ O ₃ layer.	Corrosion resistance, wear resistance, appearance. Layer can be dyed.	Aluminium architectural sections, consumer electronics housings, aerospace components
Powder Coating	Electrostatically apply dry powder then cure in oven to form hard polymer film.	Corrosion resistance, appearance. Thick, durable coating. Environmentally friendlier than liquid paint.	Steel fabrications, automotive parts, consumer goods. Excellent for outdoor structures.

6. Manufacturing Process Selection



7. Practice Exercises

#	Exercise	Scenario
1	Process identification	State the most appropriate manufacturing process for each component: (a) 50,000 identical aluminium pump housings, moderate tolerance. (b) One-off large steel bearing housing, 600mm diameter, rough tolerance. (c) High-precision shaft, diameter 50mm ± 0.005 mm, medium carbon steel. (d) 200,000 plastic electrical enclosure lids per year.
2	Casting selection	Compare sand casting and die casting for producing 10,000 aluminium motor housings. Consider: tooling cost, part cost, surface finish, dimensional tolerance, lead time. At what production volume does die casting become more economical if tooling = \$80,000 and part cost advantage = \$8 per unit?
3	Machining operation selection	For each feature, state the machining operation required and explain why: (a) A flat surface on a steel block to Ra 1.6 μ m, ± 0.02 mm flatness. (b) Six M10 threaded holes on a flange face. (c) A cylindrical shaft diameter 40mm, tolerance ± 0.01 mm. (d) A complex 3D pocket in hardened tool steel (HRC 58).
4	Welding process selection	Select and justify the welding process for each application: (a) 3mm stainless steel pressure vessel, high-quality internal finish required. (b) Structural steel portal frame erection on an outdoor construction site. (c) High volume production of 2mm carbon steel automotive body panels. (d) Repair weld on a crane hook while crane is still in service position.
5	Heat treatment specification	A medium carbon steel shaft (0.45% C) requires: surface hardness HRC 58+, core toughness retained, no distortion permitted after heat treatment. Which heat treatment process would you specify and why? What surface hardening process would you choose if the component must maintain exact dimensions after treatment?

Lesson Summary

Category / Process	Select When...
Sand casting	Large, complex shapes. Any metal. Low-medium volume. Surface finish not critical.
Die casting	High volume (>10,000 parts). Non-ferrous (Al, Zn, Mg). Good surface finish needed.
Hot forging	High-strength structural parts (shafts, gears, cranks). Grain structure aligned to load path.
Rolling / extrusion	Constant cross-section profiles in long lengths. Structural sections, wire, sheet, tube.
Turning	Cylindrical parts, threads, bores. Use CNC for precision ($\pm 0.01\text{mm}$).
Milling	Flat surfaces, slots, pockets, complex contours. Most versatile machining operation.
Grinding	Tight tolerance ($\pm 0.001\text{mm}$), fine surface finish. After heat treatment where distortion occurred.
MIG welding	General fabrication, structural steel. Fast, versatile, semi-skilled operators.
TIG welding	Stainless, aluminium, thin materials, pressure vessels. Highest quality. Skilled operators.
Quench and Temper	Medium/high carbon steel requiring high strength and hardness throughout section.
Case hardening	Low carbon steel requiring hard wear-resistant surface and tough core (gears, cams).
Electroplating / anodising	Corrosion protection, appearance, wear resistance. Aluminium: anodise. Steel: zinc/chrome plate.

What Is Next — MECH-009

MECH-009: Welding Technology — Deep dive into welding metallurgy, joint design, welding symbols on drawings, destructive and non-destructive testing (NDT), weld defects and their causes, welding procedure specifications (WPS), and welder qualification. Includes structural steel and pressure vessel welding requirements per BS EN 1011 and ASME IX.

Resource	Details
MECH PDF Series	MECH-001 through MECH-030 — ayetechub.com/pdfs.html
Engineering Formulas	Free reference tool — ayetechub.com/downloads/engineering-formulas.html
Community	Telegram: t.me/ayetechub